

## BPHO ROUND 0 · QUESTION PAPER

## Drill bank section 2

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.***Before you start**

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

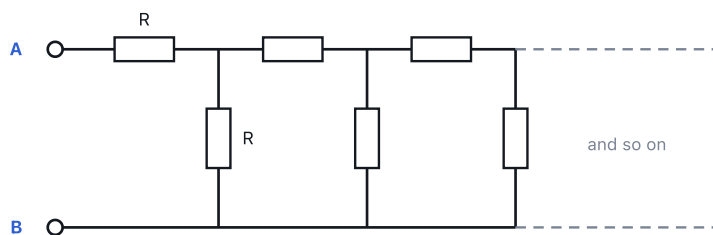
**Q1** · Dimensional analysis: the speed of a capillary ripple **A**

Very short ripples on deep water are held in shape by surface tension rather than by gravity. The speed  $v$  of such a ripple is found to depend only on the surface tension  $\gamma$  of the water, on its density  $\rho$ , and on the wavelength  $\lambda$ . Which expression is dimensionally a speed?

- A**  $\sqrt{(\gamma\rho/\lambda)}$  **B**  $\sqrt{(\gamma\lambda/\rho)}$   
**C**  $\sqrt{(\gamma/\rho\lambda)}$  **D**  $\gamma/(\rho\lambda)$   
**E**  $\sqrt{(\gamma\rho)}/\lambda$

**Q2** · An endless resistor ladder: resistance between the input terminals **H**

The network below repeats the same two-resistor section without end: a resistor  $R$  in series, then a resistor  $R$  in parallel with whatever follows.

Every resistor in the network has the same resistance  $R$ .

What is the resistance measured between the two terminals on the left?

- A**  $R(1 + \sqrt{5})/2$  **B**  $R(1 + \sqrt{3})$   
**C**  $2R$  **D**  $R\sqrt{2}$   
**E**  $R(3 + \sqrt{5})/2$

**Q3** · One wavelength on two metals: which one emits, and what the stopping potential is **L**

Light of wavelength 400 nm falls on two clean metal surfaces. Metal A has a work function of 2.0 eV and metal B has a work function of 3.5 eV.

Which statement is correct?

- A** Only A emits, and its stopping potential is 1.1 V      **B** Both metals emit, with stopping potentials of 1.1 V for A and 0.40 V for B
- C** Only B emits, with a stopping potential of 0.40 V      **D** Neither metal emits, because 400 nm is longer than either threshold wavelength
- E** Both metals emit, and the two stopping potentials are equal

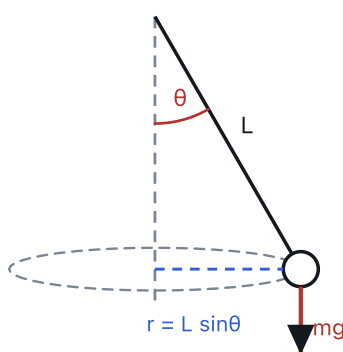
**Q4** · A bouncing ball: total time before the bouncing dies away **B**

A ball is released from rest at a height **h** above a hard floor. After each bounce it rises to exactly one quarter of the height from which it fell, and it never stops perfectly. Neglecting air resistance, what is the total time from release until the bouncing has died away?

- A**  $2\sqrt{(2h/g)}$       **B**  $3\sqrt{(2h/g)}$
- C**  $4\sqrt{(2h/g)}$       **D**  $(1 + \sqrt{2})\sqrt{(2h/g)}$
- E**  $(5/2)\sqrt{(2h/g)}$

**Q5** · Conical pendulum: the period as a function of the cone angle **D**

A small bob hangs on a light inextensible string of length **L**. The bob is set moving in a horizontal circle so that the string sweeps out a cone, making a constant angle  **$\theta$**  with the vertical.



The bob moves in a horizontal circle of radius  $r$ .  
The tension has to hold the weight up as well as turn the bob.

What is the period of the motion?

- A**  $2\pi\sqrt{(L/g)}$       **B**  $2\pi\sqrt{(L \sin \theta/g)}$
- C**  $2\pi\sqrt{(L/(g \cos \theta))}$       **D**  $2\pi\sqrt{(L \cos \theta/g)}$
- E**  $2\pi\sqrt{(g/(L \cos \theta))}$

**Q6** · A water jet striking a wall: the force from a rate of momentum loss **C**

Water of density  $\rho$  flows steadily along a pipe of internal cross-sectional area  $A$  with speed  $v$ . It leaves the pipe and strikes a flat wall head-on, coming to rest against it. What is the average force the water exerts on the wall?

**A**  $\rho Av$

**B**  $2\rho Av^2$

**C**  $\rho Av^2/2$

**D**  $\rho A^2 v$

**E**  $\rho Av^2$

---

**Q7** · Extension of a wire when both the length and the diameter are changed **E**

A wire of length  $L$  and diameter  $d$  extends by  $e$  when a load  $F$  is hung on it. A second wire of the same material has length  $2L$  and diameter  $d/2$ . What load is needed to make it extend by the same amount  $e$ ?

**A**  $F/8$

**B**  $F/4$

**C**  $F/2$

**D**  $2F$

**E**  $F$

---

**Q8** · Doppler shift: the frequency heard from a source closing at  $v/5$  **F**

A car sounding its horn at a steady frequency travels in a straight line towards a stationary observer at a steady speed of  $v/5$ , where  $v$  is the speed of sound in air. What is the ratio of the frequency the observer hears to the frequency the horn emits?

**A**  $5/4$

**B**  $4/5$

**C**  $6/5$

**D**  $25/16$

**E**  $5/6$

---

**Q9** · A prism at minimum deviation: finding the deviation from the refractive index **G**

A ray of monochromatic light passes through a glass prism of refracting angle  $60^\circ$  and refractive index  $\sqrt{2}$ . The prism is turned until the deviation is as small as it can be. What is the angle of minimum deviation?

**A**  $45^\circ$

**B**  $60^\circ$

**C**  $30^\circ$

**D**  $15^\circ$

**E**  $22.5^\circ$

---

**Q10** · Parallel resistors: which one takes the larger share of the power H

A  $6\ \Omega$  resistor and a  $3\ \Omega$  resistor are connected in parallel across a battery of negligible internal resistance. What fraction of the total power supplied by the battery is dissipated in the  $3\ \Omega$  resistor?

**A**  $1/3$

**B**  $2/3$

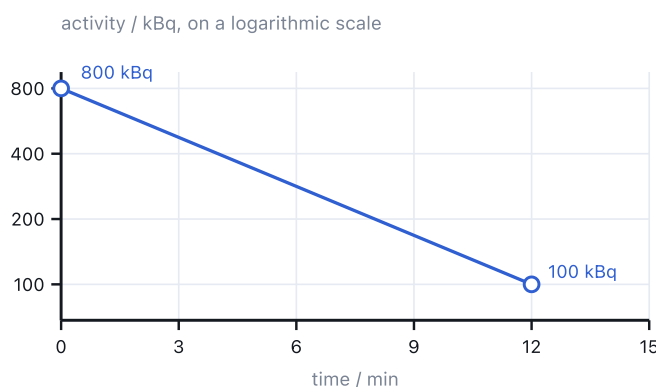
**C**  $1/2$

**D**  $4/9$

**E**  $5/6$

**Q11** · Half-life from a decay curve plotted with a logarithmic activity axis K

The activity of a pure radioactive source is measured over twelve minutes. The results are plotted with the activity on a **logarithmic** scale and the time on a linear scale.



The plotted points lie on a straight line. What is the half-life of the source?

**A** 1.5 min

**B** 3 min

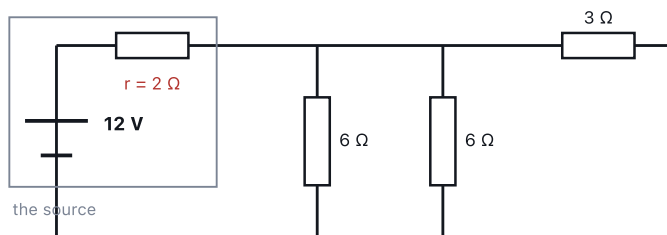
**C** 4 min

**D** 6 min

**E** 12 min

**Q12** · Power in one resistor of a network driven by a cell with internal resistance H

A cell of electromotive force  $12\text{ V}$  and internal resistance  $2\ \Omega$  is connected to two  $6\ \Omega$  resistors in parallel with each other, and that combination in series with a  $3\ \Omega$  resistor.



The source has an internal resistance of  $2\ \Omega$ .

The two  $6\ \Omega$  resistors are in parallel with each other.

What is the power dissipated in the  $3\ \Omega$  resistor?

**A**  $4.5\text{ W}$

**B**  $6.75\text{ W}$

**C**  $12\text{ W}$

**D**  $13.5\text{ W}$

**E**  $18\text{ W}$

**Q13** · A bullet embedding in a block: momentum first, then friction, then the coefficient C

A bullet of mass  $20\text{ g}$  travelling horizontally at  $300\text{ m/s}$  embeds itself in a block of mass  $1980\text{ g}$  resting on a rough horizontal surface. The block and bullet then slide  $1.5\text{ m}$  before coming to rest. Take  $g = 10\text{ m s}^{-2}$ .

What is the coefficient of friction between the block and the surface?

**A**  $0.30$

**B**  $0.60$

**C**  $0.20$

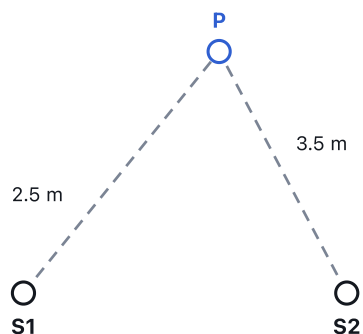
**D**  $0.15$

**E**  $0.45$



**Q16** · Two-source interference: the lowest frequency for which a given point is a minimum **F**

Two loudspeakers **S1** and **S2** are driven by the same oscillator and therefore emit sound in phase. The speed of sound in air is  $340 \text{ m s}^{-1}$ . A microphone at the point **P** is **2.5 m** from **S1** and **3.5 m** from **S2**.



S1 and S2 are driven by the same oscillator, so they emit in phase.

What is the **lowest** frequency at which the sound heard at **P** is a minimum?

**A** 85 Hz

**B** 170 Hz

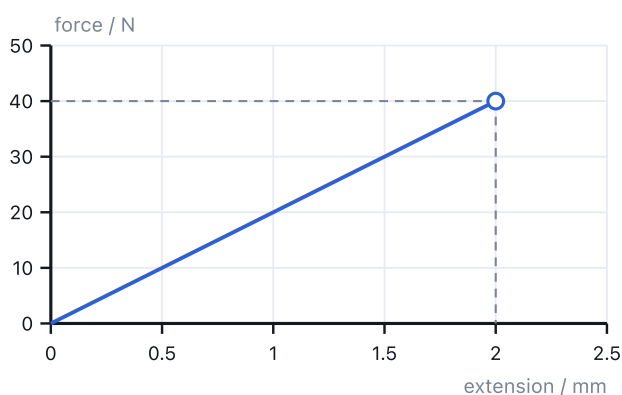
**C** 340 Hz

**D** 510 Hz

**E** 680 Hz

**Q17** · The Young modulus from a force-extension graph and the dimensions of the wire **E**

The graph shows how the extension of a wire varies with the applied force, up to the limit of proportionality. The wire has an unstretched length of **2.0 m** and a cross-sectional area of  $1.0 \times 10^{-7} \text{ m}^2$ .



What is the Young modulus of the material of the wire?

**A**  $1.0 \times 10^{11} \text{ Pa}$

**B**  $2.0 \times 10^{11} \text{ Pa}$

**C**  $4.0 \times 10^{11} \text{ Pa}$

**D**  $8.0 \times 10^{11} \text{ Pa}$

**E**  $1.6 \times 10^{12} \text{ Pa}$

**Q18** · How the number of photons emitted per second depends on power and wavelength **L**

A laser emits monochromatic light of wavelength  $\lambda$  at a power  $P$ . The wavelength is then **halved** and the power is **doubled**.

By what factor does the number of photons emitted per second change?

**A**  $1/4$

**B**  $1/2$

**C**  $1$

**D**  $2$

**E**  $4$

---

**Q19** · The first-order approximation to a series combination when one term dominates **A**

Two springs of stiffness  $k$  and  $K$  are joined end to end, so that the pair is stretched by pulling on the two free ends. The combined stiffness is

$$k_{\text{combined}} = kK/(k + K)$$

In the case where  $K$  is very much larger than  $k$ , which expression is correct **to first order** in the small ratio  $k/K$ ?

**A**  $k(1 - k/K)$

**B**  $k(1 + k/K)$

**C**  $k(1 - K/k)$

**D**  $K(1 - k/K)$

**E**  $k$

---

**Q20** · The number of images of a point object between two plane mirrors at an angle **G**

Two plane mirrors are arranged with their reflecting surfaces facing each other, meeting along a common vertical edge at an angle of  $60^\circ$ . A small illuminated bead is placed between them, well away from the edge.

How many images of the bead are formed?

**A**  $3$

**B**  $4$

**C**  $5$

**D**  $6$

**E**  $11$

---



**Q24** · An order-of-magnitude estimate of the mass of air in a room **A**

A classroom measures about 10 m by 8 m by 3 m. The density of air is about  $1.2 \text{ kg m}^{-3}$ .

Which value is the best estimate of the mass of the air in the room?

**A** 3 kg

**B** 30 kg

**C** 300 kg

**D** 3000 kg

**E** 30000 kg

---

**Q25** · The maximum kinetic energy of photoelectrons when the wavelength is halved **L**

The work function of a metal surface is 2.0 eV. Light of wavelength  $\lambda$  incident on the surface produces photoelectrons whose maximum kinetic energy is 1.0 eV.

The wavelength of the incident light is then halved. What is the new maximum kinetic energy of the photoelectrons?

**A** 1.0 eV

**B** 2.0 eV

**C** 3.0 eV

**D** 4.0 eV

**E** 6.0 eV